

PORTFOLIO SAMPLE — This document was independently authored for demonstration purposes only, based on GE Vernova's public information about g³ (g-cubed) SF6-free gas technology. It is not an official GE Vernova document and is not affiliated with, endorsed by, or sourced from GE Vernova. It does not cover any specific breaker model's real ratings, settings, or OEM procedures. Always follow the OEM instruction manual, gas safety data sheet (SDS), and site protection/isolation procedures for the specific breaker being serviced.

GE VERNOVA — GRID SOLUTIONS (PORTFOLIO SAMPLE)

TROUBLESHOOTING MANUAL

g³ SF6-Free Circuit Breaker

(g-cubed gas insulation technology)

TS-CB-G3-2211 | Revision 01 | Effective Date: 2026-07-31

Document Control

Document No.	TS-CB-G3-2211
Title	g ³ (g-cubed) SF6-Free Circuit Breaker Troubleshooting Manual
Equipment Class	HV dead tank and live tank circuit breakers using g ³ (CO2 / O2 / fluoronitrile) insulating and switching gas
Revision	01
Effective Date	2026-07-31
Audience	Qualified substation/switchgear maintenance technicians and protection engineers

1. Purpose and Scope

This manual provides a structured troubleshooting reference for high-voltage circuit breakers using g³ (“g-cubed”) insulating and switching gas — GE Vernova's SF6-free alternative used in dead tank and live tank breaker designs. g³ gas is a mixture of carbon dioxide (CO2), oxygen (O2), and a fluoronitrile additive, engineered to match SF6's dielectric and interruption performance at a much lower global warming potential.

Because g³ breakers use the same dead tank / live tank mechanical architecture as their SF6 equivalents, most mechanical, control-circuit, and operating-mechanism troubleshooting is identical to a conventional gas breaker. This manual focuses on where g³-specific behavior changes the diagnostic picture, and cross-references the general procedure for everything else.

NOTE: This manual does not replace the OEM instruction manual, nameplate ratings, or site protection/isolation procedures for the specific breaker being serviced, all of which take precedence over this document.

2. g³ vs. SF6: What Changes for Troubleshooting

g³ gas is designed as a drop-in-class replacement for SF6 in terms of equipment footprint and performance, but its physical and chemical properties differ in ways that matter during diagnosis.

Property	SF6	g ³
Composition	Pure sulphur hexafluoride gas	Mixture of CO2, O2, and a fluoronitrile additive (C4F7N / 3M Novec 4710)
Global warming potential	Very high (a potent greenhouse gas)	Approximately 99% lower than SF6 in

Property	SF ₆	g ³
		equipment gas mass
Equipment footprint	Baseline	Same dimensions and performance class as equivalent SF ₆ equipment
Cold-weather behavior	Standard SF ₆ liquefaction curve	The fluoronitrile component has a higher liquefaction temperature than SF ₆ and is more sensitive to low-temperature condensation — density/temperature compensation is critical
Decomposition byproducts	Toxic, corrosive gaseous/solid byproducts from arcing	Also produces toxic, corrosive byproducts from arcing — treat any opened compartment with the same handling caution as SF ₆ , following the OEM/SDS guidance for g ³ byproducts specifically

3. Safety Precautions

WARNING: g³ breakers store the same mechanical energy (charged springs, hydraulic/pneumatic accumulators) as conventional breakers. Never work inside a mechanism compartment without first discharging or blocking stored energy per the OEM manual.

- Do not begin work until a valid switching order/permit has been issued, the breaker is confirmed open, isolated, and grounded, and lockout/tagout is applied.
- Consult the gas supplier's Safety Data Sheet (SDS) for g³ gas and its arc decomposition byproducts before opening any gas compartment — do not assume SF₆ handling guidance applies without checking, since the byproduct chemistry differs.
- Use appropriate PPE and ventilation when opening any compartment that has experienced arcing, consistent with the OEM's g³ handling procedure.
- Be aware that g³'s fluoronitrile component has a higher liquefaction temperature than SF₆. In cold ambient conditions, verify the breaker's density reading is temperature-compensated correctly before treating a low reading as a leak.
- Use appropriate arc-flash PPE per the site's arc-flash study/label for any work near energized conductors or during switching operations.
- Never bypass a protection or interlock function without following the site's management of change (MOC) and authorized bypass process.

4. Required Tools and Test Equipment

- Multimeter and/or micro-ohmmeter (contact resistance testing)
- Circuit breaker timing analyzer (open/close timing and contact travel)
- g³-compatible gas density/pressure gauge and leak detector — confirm the instrument is rated for g³ gas composition, not SF₆-only
- Insulation resistance (megohmmeter) tester
- Trip/close coil current signature test set
- Control schematic/wiring diagram and OEM instruction manual for the specific breaker

- g³ gas Safety Data Sheet (SDS) and OEM handling/recovery procedure
- Lockout/tagout kit and grounding equipment per site procedure
- Appropriate arc-flash and electrical PPE per the site's arc-flash study

5. General Troubleshooting Methodology

1. **Gather the event data.** Record what happened: alarm/annunciation text, SCADA event log, protection relay targets, and whether the breaker was commanded to operate or failed spontaneously.
2. **Confirm breaker status.** Verify actual breaker position (open/closed) using mechanical indicators, not just control room status.
3. **Isolate and ground.** Follow the site switching order to isolate and ground the breaker before any hands-on diagnostic work requiring compartment access.
4. **Check the obvious first.** Verify control power, gas pressure/density (temperature-compensated), and any active lockout or block-close condition before assuming a component failure.
5. **Consult the relevant table.** Use the troubleshooting tables in Sections 6–9 to narrow the probable cause based on the specific symptom.
6. **Test before replacing.** Confirm a suspected component has actually failed before replacing it — misdiagnosis is a common cause of repeat failures.
7. **Document and close out.** Record the fault, root cause, and corrective action in the equipment history, and close the work permit.

6. Troubleshooting: Breaker Fails to Close

Symptom	Probable Cause	Corrective Action
No response to close command; no indication of mechanism motion	Loss of control power, blown close-circuit fuse, or open close coil	Verify control power at the cabinet; check close-circuit fuses/breakers; measure close coil resistance against OEM spec.
Mechanism attempts to close but stops partway	Low g ³ gas density has activated a block-close lockout	Check the gas density gauge/alarm status; confirm the reading is temperature-compensated before assuming a real leak; do not force a close operation.
Close command given but mechanism does not move at all (spring-operated)	Closing spring not charged, or spring-charge motor failure	Verify spring-charged indication; check spring motor control fuse and motor continuity per OEM procedure.
Close command given but anti-pump/interlock prevents closing	An active interlock is blocking the close circuit	Trace the control schematic to identify which interlock contact is open; verify the interlocking condition is actually satisfied before resetting.
Breaker closes but immediately trips ("close-and-trip")	An active protection trip condition, or a persistent fault on the protected circuit	Do not re-close without confirming the protection target and investigating the downstream circuit for an actual fault condition.

7. Troubleshooting: Breaker Fails to Trip / Open

WARNING: A breaker that fails to trip on a genuine fault condition is a critical safety and reliability issue. Escalate immediately per Section 10 if a trip failure occurs during an actual fault, rather than continuing routine troubleshooting.

Symptom	Probable Cause	Corrective Action
No response to trip command from either protection or manual trip	Loss of DC control/trip power, blown trip-circuit fuse, or open trip coil	Verify trip circuit DC supply and fuses; measure trip coil resistance against OEM spec; check trip circuit supervision relay/alarm if fitted.
Protection relay issues a trip, but breaker does not open	Failed trip coil, seized/binding mechanism, or a failed auxiliary trip contact	Perform a manual/local trip test if safe and permitted; check trip coil and auxiliary contact continuity; inspect mechanism for binding per OEM procedure.
Breaker trips slowly or with abnormal timing	Mechanism lubrication/binding issues, degraded operating springs, or a weak trip coil	Run a timing test and compare to OEM specification; inspect and service the mechanism per the OEM maintenance schedule if timing is out of tolerance.
Local manual trip works but remote/protection trip does not	Wiring or contact fault specific to the remote trip path (relay output contact, wiring, terminal)	Trace the remote trip path on the schematic; check the protection relay's trip output contact and associated wiring/terminals.

8. Troubleshooting: g³ Gas System

Symptom	Probable Cause	Corrective Action
Low gas density alarm (Stage 1)	Slow gas leak, or a temperature-compensation error at low ambient temperature	Confirm the density monitor's temperature compensation is functioning correctly before assuming a leak, given g ³ 's higher liquefaction point than SF ₆ ; log and trend the reading; schedule a leak survey.
Low gas density lockout (Stage 2 / block-close)	Gas density has dropped below the safe operating threshold for interruption	Do not attempt to operate the breaker; treat as urgent; isolate, investigate the leak source, and restore gas per OEM procedure and g ³ -rated equipment before any operation.
Audible gas leak or visible frost/condensation at a fitting	Failed seal, loose fitting, or valve leak	Follow the g ³ SDS handling/PPE guidance; use a leak detector confirmed compatible with g ³ 's mixed-gas composition (an SF ₆ -only detector may not respond correctly) to locate the leak.
Unexpected density reading swings with ambient temperature	Fluoronitrile component partially condensing at low temperature, a known g ³ characteristic if the compartment is outside its rated temperature range	Compare against the OEM's g ³ temperature/density curve for the specific breaker; confirm the installation's minimum ambient rating is appropriate for the site's climate.
Moisture/dew point alarm	Moisture ingress, often from a	Follow OEM procedure for gas analysis

Symptom	Probable Cause	Corrective Action
	compromised seal or desiccant saturation	specific to g ³ ; do not disregard — moisture increases the risk of internal flashover and byproduct formation.

9. Troubleshooting: Operating Mechanism and Control Circuit

Symptom	Probable Cause	Corrective Action
Spring-charge motor runs continuously or fails to stop	Failed limit switch or charge-indication contact	De-energize the motor circuit per LOTO; inspect and test the charging limit switch per OEM procedure.
Hydraulic mechanism low-pressure alarm	Internal leak, pump/motor fault, or temperature-related pressure drop	Check pump run frequency and pressure trend; investigate for leaks per OEM procedure before topping up fluid.
“Control circuit faulty” or supervision alarm active	An open circuit somewhere in the trip or close supervision loop	Trace the supervision circuit on the schematic; check terminal connections and continuity segment by segment.
Breaker position indication disagrees with actual mechanical position	Failed or misaligned auxiliary switch (52a/52b contacts)	Visually verify actual mechanical position first; inspect and adjust/replace the auxiliary switch per OEM procedure.

10. Escalation Criteria

Stop routine troubleshooting and escalate to engineering, the OEM, or a specialized service provider when any of the following apply:

- A breaker fails to trip during an actual fault condition (safety-critical failure).
- Gas density has reached a lockout-level low reading and the leak source cannot be quickly identified and isolated.
- Internal arcing, unusual odor, discoloration, or visible byproduct residue is found inside a gas compartment.
- Density readings behave inconsistently with the OEM's g³ temperature-compensation curve and the cause cannot be resolved by instrument verification alone.
- Timing test results are significantly out of tolerance and the cause is not resolved by routine adjustment/lubrication.
- Gas recovery, handling, or disposal is required — g³ gas recovery follows OEM- and regulation-specific procedures distinct from SF₆ recovery equipment and should not be improvised.
- Root cause cannot be isolated after following the applicable tables in this manual, and the breaker remains out of service.

11. Revision History

Rev.	Date	Author	Description of Change
01	2026-07-31	M. Langley, Tech. Writer	Initial release adapting the standard breaker troubleshooting structure to g ³ SF ₆ -free gas insulation.